

# Satyansh Dubey

+91-7890719366 | dubeysatyansh4@gmail.com | [LinkedIn](#) | [GitHub](#)

## SUMMARY

Full-Stack Software Engineer building AI-integrated, production-grade systems across React, Next.js, and TypeScript on the frontend, and FastAPI, REST APIs, and secure authentication (JWT, RBAC) on the backend. I architect autonomous agents, real-time data pipelines, and privacy-preserving systems – from sandboxed execution environments to role-based access control and cryptographic protocols.

## EDUCATION

|                                                  |                                |
|--------------------------------------------------|--------------------------------|
| <b>Heritage Institute of Technology</b>          | Kolkata, India                 |
| Bachelor of Technology in Information Technology | Aug 2024 – Jun 2028 (Expected) |
| <b>Shree Jain Vidyalaya</b>                      | Kolkata, India                 |
| Class 12 – 93%                                   | Mar 2022 – Jun 2024            |

## TECHNICAL SKILLS

|                               |                                                                                   |
|-------------------------------|-----------------------------------------------------------------------------------|
| <b>Languages:</b>             | C, C++, Python, JavaScript, TypeScript                                            |
| <b>Frontend:</b>              | React 19, Next.js 14/16, Tailwind CSS, Framer Motion, HTML, CSS                   |
| <b>Backend:</b>               | FastAPI, Node.js, Express.js, REST APIs, JWT Authentication, SQLAlchemy ORM, RBAC |
| <b>Databases:</b>             | PostgreSQL, MongoDB, SQLite, Firebase, Supabase                                   |
| <b>Tools &amp; Platforms:</b> | Git, GitHub, Docker, AWS, Playwright, Chrome MV3, Linux, Vite                     |
| <b>Core:</b>                  | Data Structures & Algorithms, System Design, OOP, Competitive Programming         |

## CERTIFICATIONS

- Oracle Database SQL Certified Associate (1Z0-071) – Oracle
- Full-Stack Developer Professional Certificate – Meta, Coursera

## PROJECTS

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |      |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| <b>PLUTO – Autonomous Cyber Defense Agent</b> — <i>Next.js 16, TypeScript, Playwright, Chrome MV3</i>                                                                                                                                                                                                                                                                                                                                                                                       | 2025 |
| GitHub <a href="#">Live Demo</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                            |      |
| <ul style="list-style-type: none"><li>• Architected an autonomous cyber defense agent using a continuous Observe-Reason-Decide-Act loop for real-time threat detection, classifying 11+ attack vectors including DDoS, brute-force, phishing, and CSP violations.</li><li>• Built a Playwright-based sandboxed interrogation engine and a Chrome MV3 extension with live telemetry, CLI monitoring, and automated mitigation (IP blocking, rate-limiting) with full audit trails.</li></ul> |      |
| <b>SmartLogistics – AI-Powered Supply Chain Platform</b> — <i>Next.js 14, Express.js, Supabase, TypeScript</i>                                                                                                                                                                                                                                                                                                                                                                              | 2025 |
| GitHub <a href="#">Live Demo</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                            |      |
| <ul style="list-style-type: none"><li>• Engineered a real-time logistics platform with RBAC-based dashboards for dispatchers, executives, and clients, backed by Express.js and Supabase (PostgreSQL).</li><li>• Built a custom AI routing engine using live weather and transit-risk data with automated API key rotation, and WebSocket-based real-time shipment tracking.</li></ul>                                                                                                      |      |
| <b>MAITRI – Offline AI Mental Health Companion</b> — <i>React, Python, Computer Vision, Voice AI</i>                                                                                                                                                                                                                                                                                                                                                                                        | 2025 |
| GitHub Profile                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |      |
| <ul style="list-style-type: none"><li>• Built a fully offline mental health companion for astronauts and high-stress professionals, combining real-time facial emotion recognition with voice-based sentiment analysis.</li><li>• Designed a sentiment-driven intervention engine and an Emotional Intelligence Briefing (EIB) system for periodic wellness reporting.</li></ul>                                                                                                            |      |
| <b>Rural Roots – AI Crop Disease Detection</b> — <i>Python, FastAPI, Computer Vision, IoT</i>                                                                                                                                                                                                                                                                                                                                                                                               | 2024 |
| <a href="#">Live Demo</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |      |
| <ul style="list-style-type: none"><li>• Developed an AI diagnostic platform using computer vision to classify crop diseases from leaf imagery, fused with IoT sensor telemetry for higher prediction accuracy.</li><li>• Built geotagged outbreak mapping to visualize and track crop disease spread across rural farming regions.</li></ul>                                                                                                                                                |      |

## ACHIEVEMENTS

- **Smart India Hackathon (SIH) 2025** – Top 5 of 500+ Teams, ISRO Ahmedabad
- **Fin Conclave HR Competition** – 1st Place
- **Hackstorm** – 1st Place
- **CIVIC AI Track** – 3rd Place (built with PLUTO)
- **Agentic AI Premier League** – 1st Runner-up
- **IDEATEX by GDG HITK** – 1st Runner-up
- **MUSE (UI/UX Design Event)** – 2nd Runner-up
- **HackAmong Us** – Top 30 of 400+ Teams
- **Hack Heritage 3.0** – Top 50